

No. of Functional Features Included in the Solution

1. Introduction

The Smart Lender – Loan Eligibility Prediction System is designed with multiple functional features that enable users to predict loan eligibility efficiently. These features improve usability, automate decision-making, and provide an interactive web-based experience for both applicants and banking professionals.

2. Functional Features Overview

The Smart Lender application includes the following core functional features:

Feature ID	Functional Feature	Description
F1	Home Page	Displays project overview and navigation.
F2	Loan Prediction Form	Collects applicant details required for prediction.
F3	Input Validation	Ensures all required fields are completed with valid data.
F4	Data Preprocessing	Encodes categorical values and scales numerical inputs before prediction.
F5	Machine Learning Prediction	Uses the trained Random Forest model to predict loan eligibility.
F6	Result Display	Displays whether the loan is Approved or Rejected.
F7	Responsive User Interface	Provides a simple and user-friendly web experience.
F8	Model Loading	Loads the trained .pkl model for real-time prediction.
F9	Error Handling	Handles invalid inputs and prevents application failures.
F10	Version Control	Maintains project history using Git and GitHub.

3. Feature Workflow

User Opens Application



Home Page



Enter Applicant Details



Input Validation



Data Preprocessing



Random Forest Prediction



Display Result

4. Feature Summary

Category	Included
Web Interface	✓
Flask Backend	✓
Machine Learning Prediction	✓
Data Preprocessing	✓
Input Validation	✓
Model Integration	✓
Prediction Result Display	✓
Responsive Design	✓
Error Handling	✓
Version Control	✓

5. Benefits of Functional Features

- Simplifies the loan application process.
- Reduces manual verification effort.
- Provides instant prediction results.
- Improves decision-making accuracy.
- Enhances user experience with a clean interface.
- Supports maintainability and future enhancements.

6. Conclusion

The Smart Lender system incorporates all essential functional features required for an intelligent loan eligibility prediction application. These features work together to deliver accurate predictions, improve operational efficiency, and provide a seamless experience for users and financial institutions.